From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ALL INDIA INTEGRATED TEST SERIES

FIITJEE

JEE (Main), 2014

PART TEST – III

Time Allotted: 3 Hours **Maximum Marks: 432**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

Caution: Question Paper CODE as given above MUST be correctly marked in the answer OMR sheet before attempting the paper. Wrong CODE or no CODE will give wrong results.

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part has only one section: **Section-A**.
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 03 and 10 to 12)** contains 6 multiple choice questions which have only one correct answer. Each question carries **+8 marks** for correct answer and **– 2 mark** for wrong answer.

Section-A (04 to 09 and 13 to 30) contains 24 multiple choice questions which have only one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

PART – I

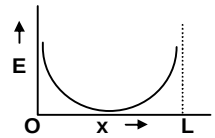
SECTION – A

Single Correct Choice Type

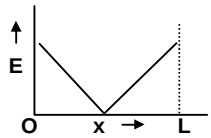
This section contains **30 multiple choice questions**. Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE is correct**.

1. Two identical point charges are placed at a separation of L. P is a point on the line joining the charges, at a distance x from any one charge. The field at P is E. E is plotted against x for values of x from close to zero to slightly less than L. Which of the following best represents the resulting curve?

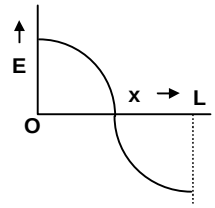
(A)



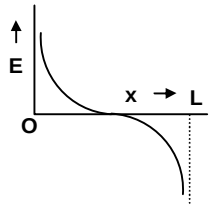
(B)



(C)

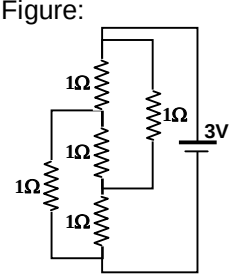


(D)

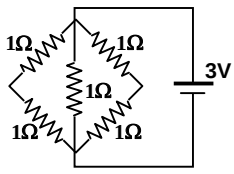


2. Figure shows three resistor configurations R1, R2 and R3 connected to 3V battery. If the power dissipated by the configuration R1, R2 and R3 is P1, P2 and P3, respectively, then

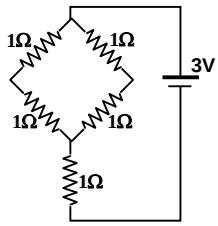
Figure:



R1



R2



R3

(A) P1 > P2 >P3
(B) P1 > P3 >P2
(C) P2 > P1 >P3
(D) P3 >P2 > P1

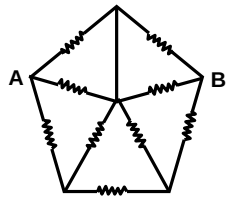
Space for rough work

3. A large sheet carries uniform surface charge density σ. A rod of length 2ℓ has a linear charge density λ on one half and $-\lambda$ on the other half. The rod is hinged at mid point O and makes θ with the normal to the sheet. The torque experienced by the rod is

(A) $\frac{\sigma\lambda\ell^2}{2\varepsilon_0}\cos\theta$ (B) $\frac{\sigma\lambda\ell}{2\varepsilon_0}\cos^2\theta$

(C) $\frac{\sigma\lambda\ell^2}{2\varepsilon_0}\sin\theta$ (D) $\frac{\sigma\lambda\ell}{2\varepsilon_0}\sin^2\theta$

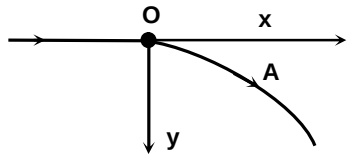
4. The effective resistance between A & B of the shown network. Where resistance of each resistor is R, is



(A) $\frac{8R}{11}$ (B) $\frac{6R}{11}$

(C) $\frac{6R}{5}$ (D) none of these

5. A positively charge moving along the positive x-direction with a speed v traces a non-linear trajectory OA in the x-y plane after passing O as shown in the figure. The combination of electric ($\vec{E}$) and magnetic field ($\vec{B}$) fields that would lead to the trajectory OA is (a, b, c are non-zero positive constants),



(A) $\vec{E} = 0;\ \vec{B} = a\hat{i} + c\hat{k}$ (B) $\vec{E} = a\hat{i};\ \vec{B} = b\hat{j} + c\hat{k}$

(C) $\vec{E} = 0;\ \vec{B} = b\hat{j} + c\hat{k}$ (D) $\vec{E} = a\hat{i};\ \vec{B} = b\hat{i} - c\hat{k}$

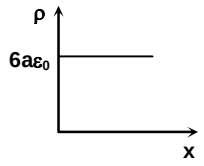
6. Current flows through a straight, thin-walled tube of radius r. The magnetic field at a distance x from the axis of the tube has magnitude B.

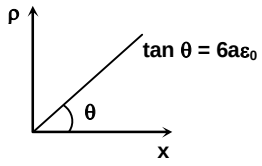
(A) $B \propto x$, for $0 < x < r$ (B) $B \propto 1/x$, for $0 < x < r$

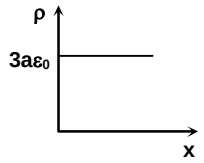
(C) $B = 0$, for $0 \le x < r$ (D) $B = 0$, only for $x = 0$

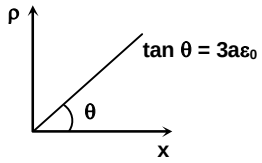
Space for rough work

7. The electric field in a certain region of space depends only on the x-coordinate as $E = 3ax^2$, where a is a constant. Then choose the correct curve, which is giving the variation of volume charge density (ρ) in the region as a function of x.

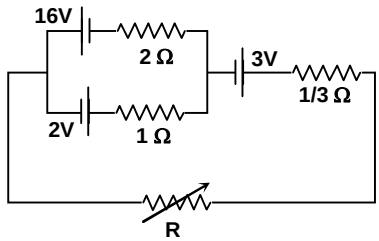
(A) 

(B) 

(C) 

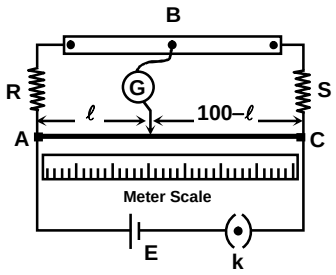
(D) 

8. Three cells of emf 16V, 2V and 3V and internal resistance 2Ω, 1 Ω and ½ Ω respectively are connected with an unknown resistance R as shown in the figure. What is the value of R such that power developed through R is maximum in steady state condition.



(A) ½ Ω
(B) 1Ω
(C) 2 Ω
(D) Cannot be determined

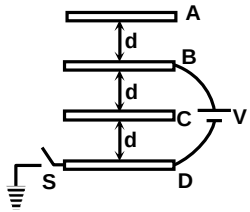
9. In a meter bridge, the null point is found at a distance of 40 cm from A. If now a resistance of 12 Ω is connected in parallel with S, the null point occurs at 60 cm from A. Determine the value of R.



(A) 15 Ω (B) 12 Ω
(C) 10 Ω (D) 5 Ω

Space for rough work

10. Find the charge flown when the switch S is closed.



(A) zero

(B) $\frac{\varepsilon_0 A}{d} V$

(C) $\frac{3}{2}\frac{\varepsilon_0 A}{d} V$

(D) $\frac{\varepsilon_0 A}{2d} V$

11. A particle of mass m and charge q is located midway between two fixed charged particles each having a charge q and at a distance 2L apart so that distance of this charge from each charge located at end is L. Assuming that the middle charge moves along the line joining the fixed charges. The frequency of oscillation when it is displaced slightly is

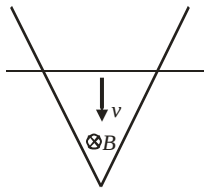
(A) $\frac{q}{\pi\sqrt{m\pi\varepsilon_0 L^3}}$

(B) $\frac{q}{4\pi\sqrt{m\pi\varepsilon_0 L^3}}$

(C) $\frac{2q}{\pi\sqrt{m\pi\varepsilon_0 L^3}}$

(D) $\frac{q}{2\pi\sqrt{m\pi\varepsilon_0 L^3}}$

12. A wire is bent in form of a V shape and placed in a horizontal plane. There exists a uniform magnetic field B perpendicular to the plane of the wire. A uniform conducting rod starts sliding over the V shaped wire with a constant speed v as shown in the figure. If the wire has uniform resistance, the current in the rod will



(A) Increase with time

(B) Decrease with time

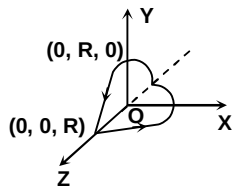
(C) Remain constant

(D) Always be zero.

13. In non-uniform magnetic field in certain direction, current loop is placed
(A) net force and net torque on loop is zero.
(B) net force may be zero but net torque is non zero.
(C) net torque may be zero but net force is non zero.
(D) neither net force nor net torque is zero.

Space for rough work

14. A circular coil of radius R is folded across its diameter such that the planes of two half rings are perpendicular to each other as shown in the figure. If the current through the coil is I, calculate the magnetic dipole moment of the configuration so formed.



(A) $\frac{\pi R^2 I}{2}(\hat{i}+\hat{j})$ (B) $\frac{\pi R^2 I}{2}(\hat{i}+\hat{k})$

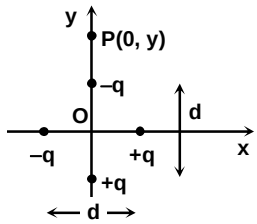
(C) $\frac{\pi R^2 I}{2}(\hat{j}+\hat{k})$ (D) $\frac{\pi R^2 I}{4}(\hat{i}+\hat{j})$

15. A particle of mass m and charge q moves at high speed along the x-axis. It is initially near $x = -\infty$ and it ends up near $x = +\infty$. A second charge Q is fixed at the point x = 0, y = –d. As the moving charge passes the stationary charge, its x component of velocity does not change appreciably, but it acquires a small velocity in y-direction. Determine the angle through which the moving charge is deflected

(A) $\theta = \tan^{-1}\left(\frac{qQ}{2\pi\varepsilon_0 dmv^2}\right)$ (B) $\theta = \sin^{-1}\left(\frac{qQ}{2\pi\varepsilon_0 dmv^2}\right)$

(C) $\theta = \tan^{-1}\left(\frac{qQ}{4\pi\varepsilon_0 dmv^2}\right)$ (D) $\theta = \sin^{-1}\left(\frac{qQ}{4\pi\varepsilon_0 dmv^2}\right)$

16. Two positive and two negative charges are kept in x-y plane in free space as shown in the figure. The magnitude of electric field due to the system of charges at a point P(0, y) will be ($y \gg d$)

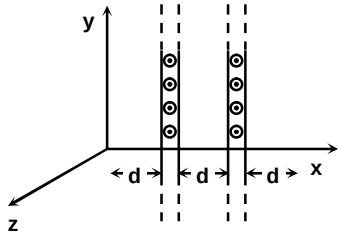


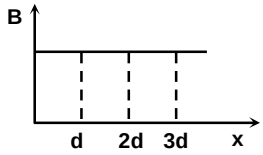
(A) $\frac{\sqrt{5}qd}{4\pi\varepsilon_0 y^3}$ (B) $\frac{2dq}{4\pi\varepsilon_0 y^3}$

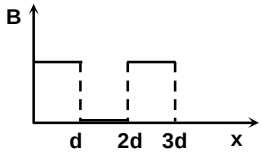
(C) $\frac{dq}{4\pi\varepsilon_0 y^3}$ (D) $\frac{3dq}{4\pi\varepsilon_0 y^3}$

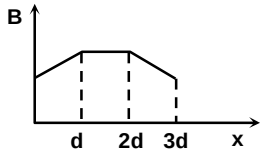
Space for rough work

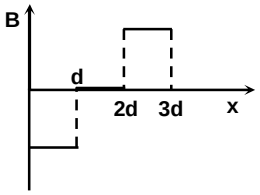
17. Two large conducting planes perpendicular to x-axis are placed at (d, 0) and (2d, 0) as shown in the figure. Current per unit width in both the planes is same and current is flowing in the outward direction. The variation of magnetic field as function of $x(0 \le x \le 3d)$ is best represented by



(A) 

(B) 

(C) 

(D) 

18. The current density in a wire of radius a varies with radial distance r as $J = \frac{J_0 r^2}{a}$, where J_0 is a constant. Choose the incorrect statement.

(A) Total current passing through the cross-section of the wire is $I = \frac{\pi J_0 a^3}{2}$

(B) Total current passing through the cross-section of the wire is $I = \frac{3\pi J_0 a^3}{2}$

(C) The field at a distance $r > a$ is $B = \frac{\mu_0 J_0 a^3}{4r}$

(D) The field at a distance $r < a$ is $B = \frac{\mu_0 J_0 r^3}{4a}$

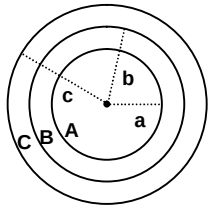
19. A washer is made of metal having resistivity $10^{-7}\Omega$M. The washer has inner radius 1 cm, outer radius 3 cm and thickness 1 mm. A magnetic field, oriented normal to the plane of the washer, has the time dependent magnitude $B = (2t)$ tesla/sec. Find the current (in ampere) around the washer.

(A) 2 Amp (B) 4 Amp

(C) 6 Amp (D) 8 Amp

Space for rough work

20. Three concentric charged metallic spherical shells A, B and C have radii a, b and c; charge densities σ, $-\sigma$ and σ; and potentials V_A, V_B and V_C respectively. Then which of the following relations is correct?

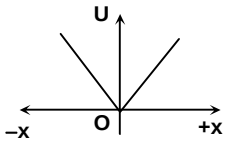


(A) $V_A = \dfrac{(a+b+c)\sigma}{\varepsilon_0}$ (B) $V_B = \dfrac{[(a^2/b)-b+c]\sigma}{\varepsilon_0}$

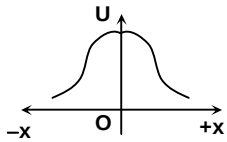
(C) $V_C = \dfrac{[(a^2+b^2)/c+c]\sigma}{\varepsilon_0}$ (D) $V_A = V_B = V_C = \dfrac{(a-b+c)\sigma}{\varepsilon_0}$

21. Four identical charges are placed at the four vertices of a square lying in YZ plane. A fifth charge is moved along X axis. The variation of potential energy (U) along X axis is correctly represented by

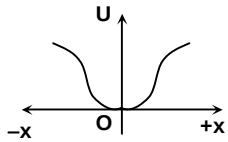
(A)



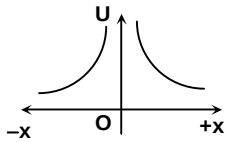
(B)



(C)



(D)

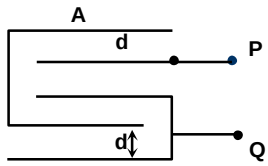


22. Two resistances 300 ohm and 400 ohm are connected in series with a 60 volt supply. A voltmeter connected across 400 ohm reads 30V. The same voltmeter when connected across the other resistance will read.

(A) 22.5 V (B) 50 V

(C) 30 V (D) 10 V

23. The equivalent capacitance between P and Q in the figure is (area of each plate is A and separation between two consecutive plates is d)



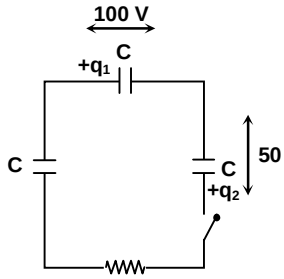
(A) $\dfrac{A\varepsilon_0}{2d}$ (B) $\dfrac{3A\varepsilon_0}{2d}$

(C) $\dfrac{2A\varepsilon_0}{d}$ (D) $\dfrac{5}{3}\dfrac{A\varepsilon_0}{d}$

Space for rough work

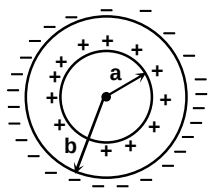
24. Two long parallel straight conductors carry currents i_1 and i_2 ($i_1 > i_2$). When the current are in the same direction, the magnetic field at a point midway between the wires is 20 μT. If the direction of i_2 is reversed, the field becomes 50 μT. the ratio of the currents i_1/i_2 is
(A) 5/2 (B) 7/3
(C) 4/3 (D) 5/3

25. Two capacitor each of capacitance C = 1 μF each charged to potential 100 V and 50 V respectively and connected across a uncharged capacitor 'C' = 1μF as shown in the figure if switch 'S' is closed then charge in the uncharged capacitor at equilibrium will be



(A) $\frac{50}{3}\mu C$ (B) 50 μC
(C) 25 μC (D) $\frac{100}{3}$ μC

26. There are two concentric spherical shells with inner shell having a positive charge q and outer shell having a charge –q. Now outer shell is supplied a charge –2q and inner shell is earthed. Calculate the ratio of initial and final capacities of the system. (If a = 3R, b = 5 R)



(A) 3/5 (B) 3
(C) 1 (D) 5/3

27. An infinite collection of current carrying conductor each carrying a current I outward perpendicular to paper are placed at x = a, x = 3a, x = 5a, ... ∞ on the x-axis. Another infinite collection of current carrying conductor each carrying current I inward perpendicular to paper are placed at x = 2a, x = 4a x = 6a, ... ∞ where a is a positive constant. Then the magnetic field at the origin due to the above collection of current carrying conductor is
(A) zero (B) $\frac{\mu_0 I}{4\pi a \ln 2}$
(C) ∞ (D) $\frac{\mu_0 I \ln 2}{2\pi a}$

Space for rough work

28. A number of resistors R_1, R_2, R_3, . . . are connected in series such that R_s is the equivalent resistance of series combination. A current I is flowing in the circuit due to a potential V applied across the circuit. V_1, V_2, V_3 . . . . are potential difference across R_1, R_2, R_3, . . . respectively. Choose the incorrect statement.

(A) Same current I will flow through each resistor

(B) $V_1 + V_2 + V_3 + \ldots = V$

(C) $V_1 = \left(\frac{R_1}{R_S}\right)V$, $V_2 = \left(\frac{R_2}{R_S}\right)V$, $V_3 = \left(\frac{R_3}{R_S}\right)V$ . . . .

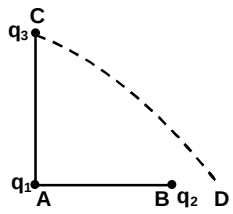
(D) $V_1 = \left(\frac{R_S}{R_1}\right)V$, $V_2 = \left(\frac{R_S}{R_2}\right)V$, $V_3 = \left(\frac{R_S}{R_3}\right)V$ . . . .

29. A soap bubble of radius R = 3 cm and thickness $t = 10^{-2}$ mm is charged to a potential of V = 0.3 volt. The bubble burst and falls as a spherical drop. Determine the potential of the drop.

(A) 3 volts. (B) 6 volts.

(C) 9 volts. (D) 12 volts.

30. Two charges q_1 and q_2 are placed 30 cm apart, as shown in the figure. A third charge q_3 is moved along the arc of a circle of radius 40 cm from C to D. The change in the potential energy of system is $\frac{kq_3}{4\pi\varepsilon_0}$, then the value of k is



(A) $8q_2$ (B) $8q_1$

(C) $6q_2$ (D) $6q_1$

Rough work

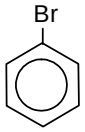
Chemistry PART – II

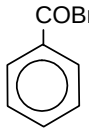
SECTION – A

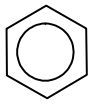
Straight Objective Type

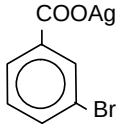
This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. Silver benzoate reacts with bromine in acetone to form

(A) 

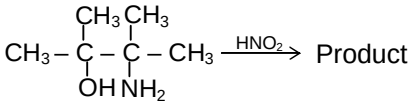
(B) 

(C) 

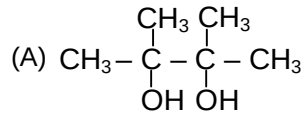
(D) 

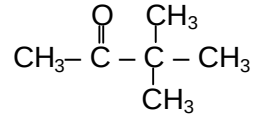
2. $RCOO^{-}Ag^{+} \xrightarrow{Br_2 / CCl_4} R-Br + CO_2 + AgBr$, This is Borodine – Hunsdiecker reaction, Reaction intermediate in this reaction is
(A) Free radical (B) Carbocation
(C) Carbene (D) Carbanion

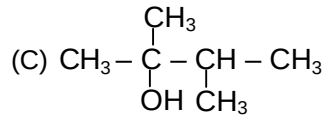
3. The product of the given reaction,

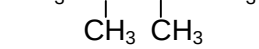


is

(A) 

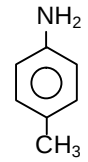
(B) 

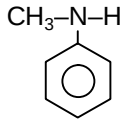
(C) 

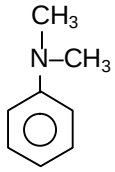
(D) 

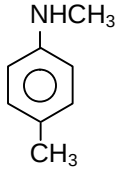
Space for rough work

4. Which of the following amine will show carbylamine reaction?

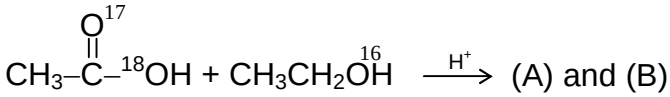
(A) 

(B) 

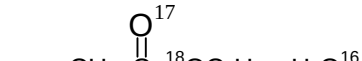
(C) 

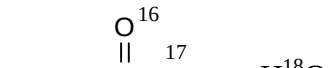
(D) 

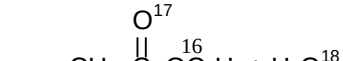
5. Consider the reaction,



(A) and (B) respectively are

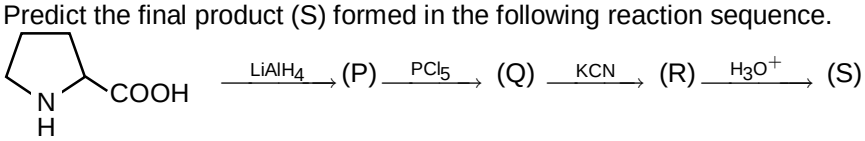
(A) 

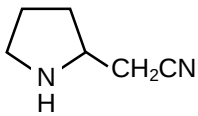
(B) 

(C) 

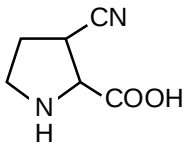
(D) Both (A) and (B)

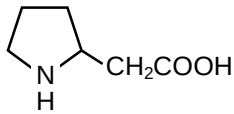
6. Predict the final product (S) formed in the following reaction sequence.



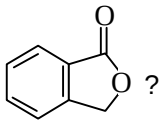
(A) 

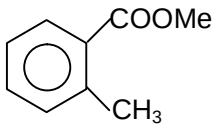
(B) $H_2N(CH_2)_3CH_2CH_2COOH$

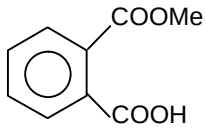
(C) 

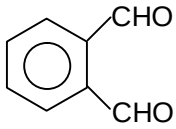
(D) 

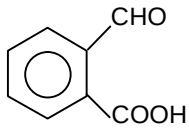
Space for rough work

7. Which of the following compound on treatment with aqueous alkali followed by acidification gives the lactone  ?

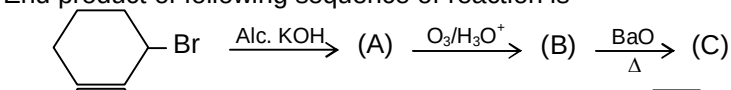
(A) 

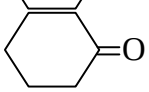
(B) 

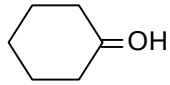
(C) 

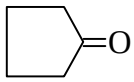
(D) 

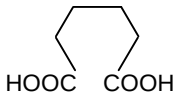
8. End product of following sequence of reaction is



(A) 

(B) 

(C) 

(D) 

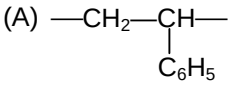
9. Propyne and propene can be distinguished by
(A) conc. H_2SO_4 (B) Br_2 inCCl_4
(C) $AgNO_3$ [Ammonical] (D) dil. $KMnO_4$

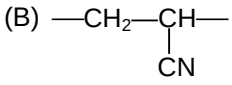
10. Identify the unknown products (A) and (B) in the given reaction:

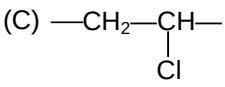
Glucose + $3C_6H_5NH–NH_2 \xrightarrow{H^{\oplus}/\Delta}$ Osazone + (A) + (B)

(A) $C_6H_5NH_2$ and NH_3 (B) $C_6H_5NH_2$ and NH_2OH
(C) C_6H_5NH—NHOH and NH_3 (D) NH_2OH and H_2O

11. The repeat unit of polystyrene is

(A) 

(B) 

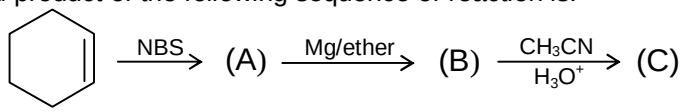
(C) 

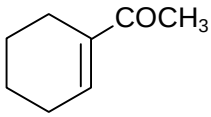
(D) —CCl_2—CCl_2—

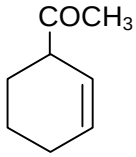
Space for rough work

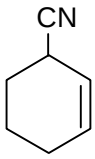
12. Buna–S rubber is a polymer of
(A) 1, 3-butadiene and styrene (B) vinyl acetate
(C) Acrylonitrile (D) none of these

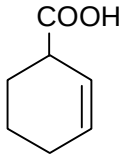
13. End product of the following sequence of reaction is:

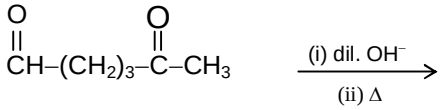


(A) 

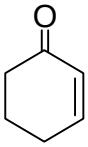
(B) 

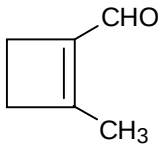
(C) 

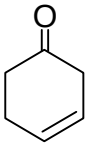
(D) 

14. 

(A), compound (A) is

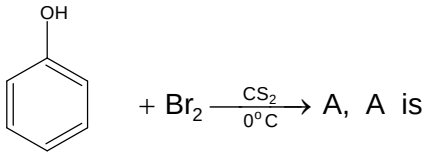
(A) 

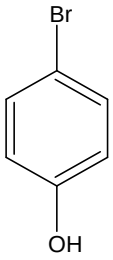
(B) 

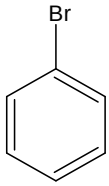
(C) 

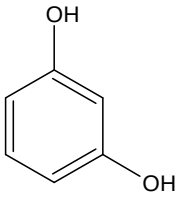
(D) None

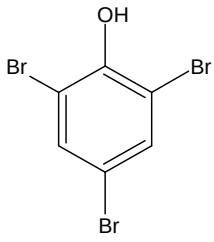
Space for rough work

15. 

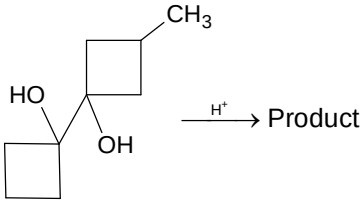
(A) 

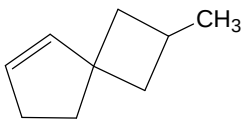
(B) 

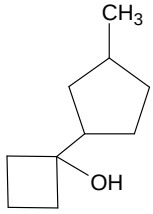
(C) 

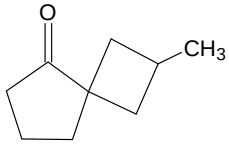
(D) 

16. The product in the following reaction is



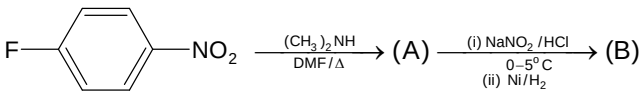
(A) 

(B) 

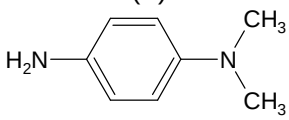
(C) 

(D) None of the above

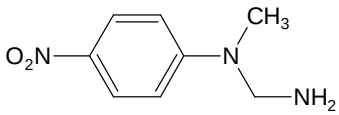
Space for rough work

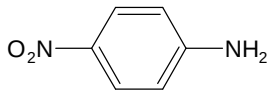
17. 

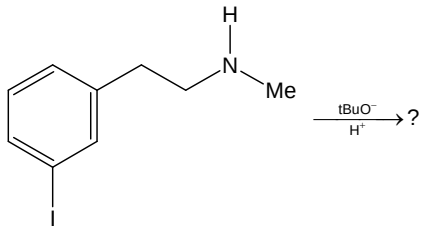
The structure of (B) is

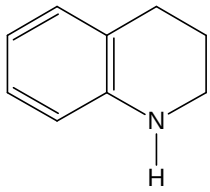
(A) 

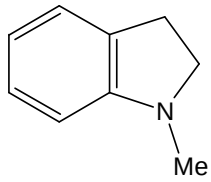
(B) 

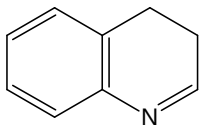
(C) 

(D) 

18. 

(A) 

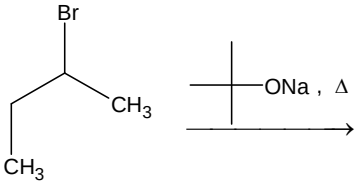
(B) 

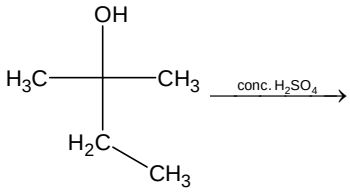
(C) 

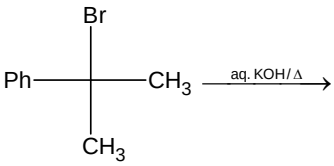
(D) All

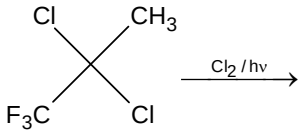
Space for rough work

19. In which reaction, product formation occurs by Saytzeff rule?

(1) 

(2) 

(3) 

(4) 

Select the correct alternative from the codes given below:

Codes:

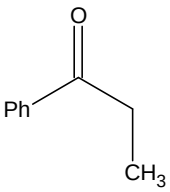
(A) 1, 2 and 3 (B) 2 and 3

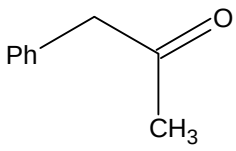
(C) 1, 2 and 4 (D) 2, 3 and 4

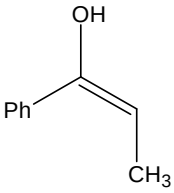
20. $Ph-C\equiv C-CH_3 \xrightarrow[H_2O]{Hg^{2+}/H^+} (W)$

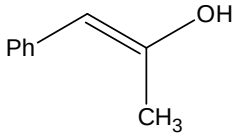
(Major)

(W) is

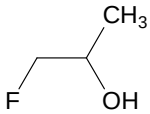
(A) 

(B) 

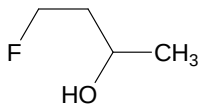
(C) 

(D) 

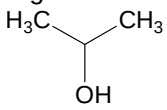
21. The order of reactivity of the following alcohols towards conc. HCl is



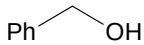
I



II



III



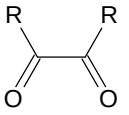
IV

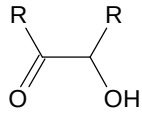
(A) I > II > III > IV (B) I > III > II > IV

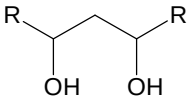
(C) IV > III > II > I (D) IV > III > I > II

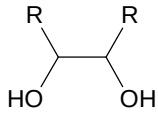
Space for rough work

22. Which of the following will be oxidized by HIO_4?

(1) 

(2) 

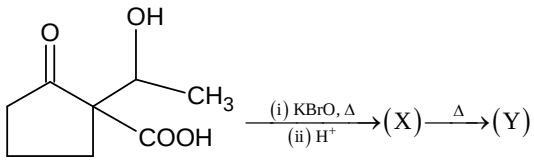
(3) 

(4) 

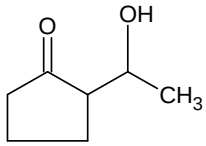
Select the correct answer using the codes below:

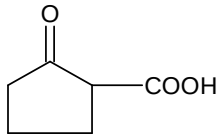
(A) 1, 2 and 3 (B) 1, 3 and 4

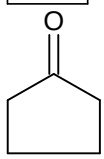
(C) 1, 2 and 4 (D) 2, 3 and 4

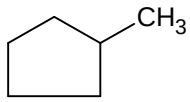
23. 

Hence the product (Y) in the above sequence of reaction is

(A) 

(B) 

(C) 

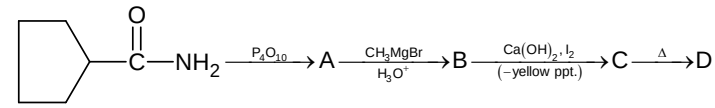
(D) 

24. Osazone formation is used to characterize

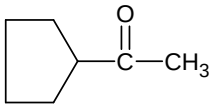
(A) polymers (B) sugars

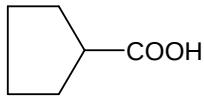
(C) carboxylic acid (D) alcohols

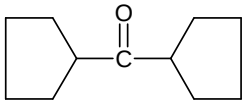
Space for rough work

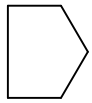
25. 

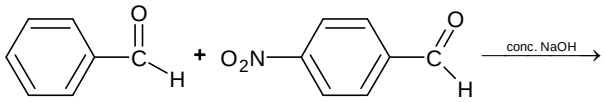
D is

(A) 

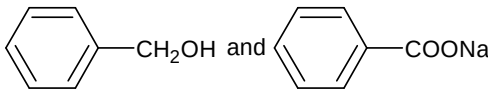
(B) 

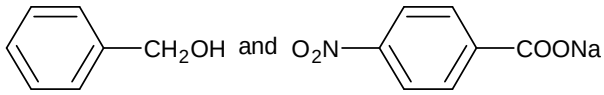
(C) 

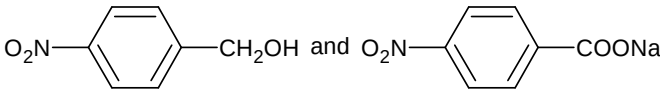
(D) 

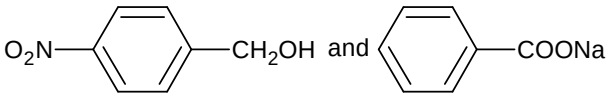
26. 

The products likely to be obtained are

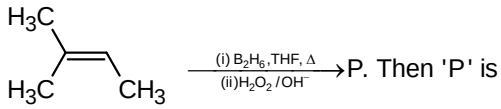
(A) 

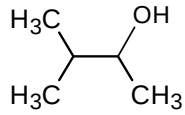
(B) 

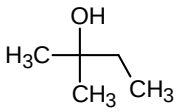
(C) 

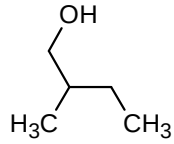
(D) 

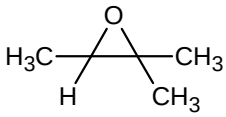
Space for rough work

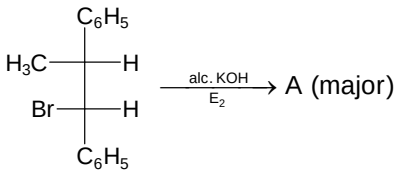
27. 

(A) 

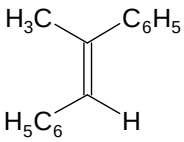
(B) 

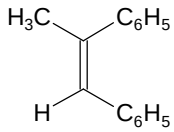
(C) 

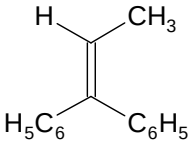
(D) 

28. 

A is

(A) 

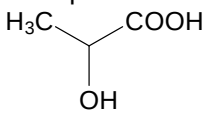
(B) 

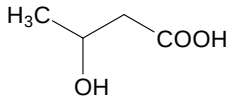
(C) 

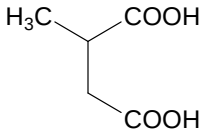
(D) All of the above

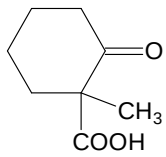
Space for rough work

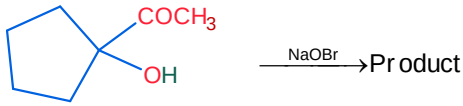
29. Which compound will lose its chirality when it is heated?

(A) 

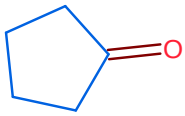
(B) 

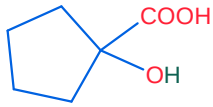
(C) 

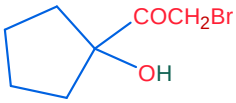
(D) 

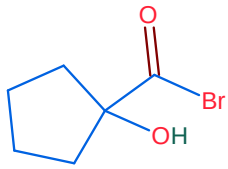
30. 

The product is

(A) 

(B) 

(C) 

(D) 

Space for rough work

Mathematics PART – III

SECTION – A

Straight Objective Type

This section contains 30 multiple choice questions numbered 1 to 30. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct.

1. If f(2010) = $\pi/2$,f(x) = f(2x) $\forall x \in R$ and f(x) is continuous for $x \in R$, then the value of $\lim_{x\to 0} \frac{\cos(f(x)) - \sqrt[3]{\cos(f(x))}}{\sin^2 x}$ is

(A) 1/4 (B) 1
(C) 0 (D) none of these

2. $\left\| \left[\int_2^3 \frac{dx}{\sqrt{1+x^3}} \right] \right\|$ is equal to (where [.] denotes the greatest integer function, |.| denotes absolute value function)

(A) 0 (B) 1
(C) 2 (D) 3

3. If $\int \frac{x\tan^{-1}x}{(1+x^2)^{3/2}}dx = \frac{k_1 \tan^{-1}x - k_2}{\sqrt{1+x^2}} + c$

where k_1, k_2 are polynomial function of variable x and c is any real constant then value of $\int_0^2 \max\{k_1, k_2\}dx$ is

(A) $\frac{-2}{3}$ (B) $\frac{2}{3}$
(C) $\frac{-3}{2}$ (D) $\frac{-1}{2}$

4. The domain of the function $\sqrt{1-f(x)}$ where $f^3(x) = 1 - x^3 - 3xf(x)$. $x \in R$ is $(f(1) \neq -1)$

(A) R (B) R – {0}
(C) $R^+ \cup \{0\}$ (D) R^-

Rough work

5. The left hand derivative of $f(x) = \begin{cases} |x-a|([x]-x), & x \neq a \\ 0 & x = a \end{cases}$ at x = a where a is an integer and [.] represent greatest integer function.

(A) $(1-a)^{-1}(-1)^{1-a}$ (B) $(1-a)^{1}(-1)^{1-a}$

(C) $(1-a)^{-1}(-1)$ (D) 1

6. $f : R \to R$; $f(x) = \sin x + x$, then the value of $\int_0^{\pi} \left(f^{-1}(x)\right) dx$ is equal to

(A) π^2 (B) $\frac{\pi^2}{2} - 2$

(C) $\frac{\pi^2}{2} + 2$ (D) can't be evaluated

7. If $f: R \to R$ is an invertible function such that f(x) and $f^{-1}(x)$ are symmetric about the line $y = -x$, then

(A) f(x) is odd

(B) f(x) and $f^{-1}(x)$ may not be symmetric about the line y = x

(C) f(x) may not be odd

(D) none of these

8. If $f(x + y + z) = f(x) + f(y) + f(z)$ with f(1) = 1 and f(2) = 2 and x, y, z $\in$ R, then evaluate $\lim_{n\to\infty} \frac{\sum_{r=1}^{n} (4r) f(3r)}{n^3}$ is equal to

(A) 4 (B) 1

(C) 2 (D) 5

9. $\int_0^{\frac{\pi}{4}} e^{\tan x}(\cos x + \sec x \tan x)dx = k$, then $[k^2]$ is

(A) 1 (B) 2

(C) 3 (D) 4

Rough work

10. f(x) is a polynomial function and $[f(\alpha)]^2 + [f'(\alpha)]^2 = 0$ then find $\lim_{x\to\alpha} \frac{f(x)}{f'(x)}\left[\frac{f'(x)}{f(x)}\right]$ where [.] denotes greatest integer function

(A) 0 (B) 1
(C) –1 (D) none of these

11. The shortest distance between the curve $y = e^x$ and the line $y = x$ is

(A) $\sqrt{2}$ units (B) $\frac{1}{2}$ units
(C) 2 units (D) $\frac{1}{\sqrt{2}}$ units

12. If $g(x) = \max|y^2 - xy|$ $(0 \le y \le 1)$ then minimum value of g(x) (for real x) is

(A) $\frac{1}{4}$ (B) $3 - \sqrt{8}$
(C) $3 + \sqrt{8}$ (D) $\frac{1}{2}$

13. sinax + cosax and |sinx| + |cosx| are periodic functions of same fundamental period if |a| equals

(A) 1 (B) 2
(C) 3 (D) 4

14. A polynomial P(x) with integral coefficients takes the value 5 at four distinct integral values of x. The number of integral solutions of the equation P(x) – 3 = 0 is

(A) 0 (B) 1
(C) 2 (D) 3

15. If $\int \frac{\left(x^2-1\right)}{\left(x^4+3x^2+1\right)\tan^{-1}\left(\frac{x^2+1}{x}\right)}dx = k \log\left|\tan^{-1}\frac{x^2+1}{x}\right| + c$, then k is equal to

(A) 0 (B) 1
(C) 2 (D) 3

Rough work

16. Range of $f(x) = \cos^{-1}x + 2\sin^{-1}x + 3\tan^{-1}x$ is

(A) $\left[-\frac{3\pi}{4}, \frac{7\pi}{4}\right]$ (B) $\left[-\frac{5\pi}{2}, \frac{7\pi}{2}\right]$

(C) $\left[-\frac{3\pi}{2}, \frac{5\pi}{2}\right]$ (D) none of these

17. Number of solutions of $\tan^{-1}|x| - \cos^{-1} x = 0$ is/ are

(A) 2 (B) 1

(C) 0 (D) none of these

18. $\int \frac{(x^2-1)}{(x^2+1)\sqrt{x^4+1}}dx$ is equal to

(A) $\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}\,x}\right)+c$ (B) $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}\,x}\right)+c$

(C) $\frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}}\right)+c$ (D) none of these.

19. $\int_0^{\infty}\left[\frac{3}{x^2+1}\right]dx$ is equal to ([.] denotes the G.I.F)

(A) $\sqrt{2}$ (B) $\sqrt{2}+1$

(C) $\frac{3}{\sqrt{2}}$ (D) infinite

20. The domain of the function $f(x) = \sqrt{3-2^x-2^{1-x}} + \sqrt{\sin^{-1}x}$ is

(A) (0, 1) (B) [0, 1)

(C) [0, 1] (D) (0, 1]

Rough work

21. Let f(x) be a real valued bijective function such that $f'(x) = \sin^2\{\sin(x+1)\}$ and f(0) = 3. The value of $\left(f^{-1}\right)'(3)$ will be

(A) $\frac{1}{\cos^2(\cos 4)}$ (B) 2 cos (sin 4) sin4

(C) $\frac{1}{\sin^2(\sin 1)}$ (D) $\frac{1}{\cos^2(\sin 1)}$

22. The number of inflection points of a function given by a third degree polynomial is exactly

(A) 2 (B) 1

(C) 3 (D) 0

23. The integral $\int \frac{\sin 2x\left(1-\frac{3}{2}\cos x\right)}{e^{\sin^2 x+\cos^3 x}}dx$ is equal to

(A) $e^{\sin^2 x+\cos^3 x} + c$ (B) $-e^{-\left(\sin^2 x+\cos^3 x\right)} + c$

(C) $e^{\left(\sin^2 x+\cos^{3/2} x\right)^2} + c$ (D) none of these

24. $\lim_{x\to 0} \frac{1-\cos^{12}(x^6)}{\sin x \sin 2x \sin 3x....\sin 12x}$ is equal to

(A) $\frac{1}{(12)!}$ (B) $\frac{1}{2(11)!}$

(C) $\frac{1}{72}$ (D) $\frac{1}{6\times(11)!}$

Rough work

25. $\lim_{x\to 0}\left\{\frac{\sin x}{x}\right\}^{\frac{1}{\left\{\frac{\tan x}{x}\right\}}}$ is equal to (where {x} = fractional part of x)

(A) 0 (B) $\frac{1}{e}$

(C) $\frac{1}{\sqrt{e}}$ (D) 1

26. Let $f(x)=\sqrt{ax^3+bx^2+cx+d}$ have local minima and maxima at x = –2 and x = 2 respectively. If 'a' is the root of the equation $x^2 - x - 6 = 0$, then which of the following is/are correct

(A) a = –2 or 3, b = 0,c = 24, d > 32 (B) a = –2, b = 0, c = 24, d < 32

(C) a = 3, b = 0, c = 24, d > 32 (D) a = –2, b = 0, c = 24, d > 32

27. If f(x) = g(x) + g(2b – x), $-b \le x \le 3b$ and g(x) exist $\forall x \in [-b, 3b]$, then

(A) x = b is never a critical point for f(x) (B) f(x) is symmetric about the line x = b

(C) f(x) is symmetric about the line x = 2b (D) none of these.

28. $\int_0^1 \left(1+(2008)x^{2008}\right)e^{x^{2008}}\,dx$ equals

(A) e^{-1} (B) $e^{-1/4}$

(C) $e^{1/2}$ (D) e

29. The value of the definite integral $\int_0^1 e^{-x^2}dx + \int_1^{1/e}\sqrt{-\ln x}\,dx$

(A) e^{-1} (B) $e^{-5/4}$

(C) $e^{3/2}$ (D) e^2

30. $\lim_{n\to\infty}\left(\frac{1^1\cdot 2^2\cdot 3^3.....(n-1)^{n-1}\cdot n^n}{n^{1+2+3+.....+n}}\right)^{\frac{1}{n^2}}$ equals

(A) e^{-1} (B) $e^{-1/4}$

(C) $e^{5/2}$ (D) e

Rough work